

Common Sector Relabeling as an Explicit Hypothesis

2026-05-10

1 The assertion

For a blocking period p , the physical index set of the blocked tensor is canonically identified with words of length p :

$$\text{Fin}(d^p) \simeq (\text{Fin } p \rightarrow \text{Fin } d).$$

The cyclic-sector construction uses the canonical multi-index relabeling blockwise. Write $X \sim_{\text{MPV}} Y$ when two tensors generate the same matrix product vector family at every length and every word. Write $R(A_k^{[p]})$ for the length- p blocked tensor $A_k^{[p]}$ read through this canonical word relabeling.

The mathematical assertion is that assembling the weighted block-diagonal tensor before this relabeling gives the same MPV family as assembling after the relabeling:

$$\bigoplus_k \mu_k^p \cdot A_k^{[p]} \sim_{\text{MPV}} \bigoplus_k \mu_k^p \cdot R(A_k^{[p]}).$$

Here $A_k^{[p]}$ denotes the length- p blocking of the k -th primitive block.

2 Relation to the paper

The paper arXiv:1606.00608 does not state this as a separate hypothesis. In the source argument the blocked-word identification is an automatic consequence of the canonical blocking construction: blocking a tensor and then reading the result through the explicit multi-index grouping of $\{1, \dots, d\}^p$ is definitionally the same operation. The paper therefore treats the equality as transparent and does not name it.

3 Status

No deviation from the source paper is present. The relabeling assertion is the identity between two descriptions of the same blocked-word coordinates, and the formalization proves it unconditionally before using it in the sector-comparison theorems.¹

4 Verdict

The hypothesis is therefore a named form of the paper’s implicit blocked-index identification, not an additional mathematical assumption.

References

¹The proof is `CommonGroupedBlockCastHypothesis.of.flattenWordOfBlock.cast_eq`, and `CommonGroupedBlockCastHypothesis.toRelabelingHypothesis` gives the stated relabeling hypothesis. Both declarations are in `TNLean/MPS/CanonicalForm/SectorComparison/CommonSectorTransport.lean`.